

Technical Document #682: Retrieval Augmented Generation - Comprehensive Overview

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Hybrid search combines keyword-based and semantic search. It leverages the strengths of both approaches for better retrieval quality.

Query decomposition breaks complex queries into simpler sub-queries. It improves retrieval quality for multi-faceted questions.

Evaluation metrics for RAG include faithfulness, answer relevance, and context recall. They measure the quality of the retrieval-generation pipeline.

Vector databases store dense embeddings for semantic search. Pinecone, Weaviate, and ChromaDB are popular vector database solutions.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Key Takeaways:

- Chunking splits documents into smaller segments for embedding.
- Query decomposition breaks complex queries into simpler sub-queries.
- Retrieval-Augmented Generation (RAG) combines information retrieval with text generation.